

High-Order Explicit Manufactured FDA Solver Verification Report

Generated report for `highOrderManufacturedFDAExplicit`

May 2, 2026

Abstract

This report documents the manufactured-solution finite-volume solver `highOrderManufacturedFDAExplicit`. The solver advances a monodomain-like activation equation coupled to the local state variables u_1 , u_2 , and u_3 . The spatial discretization may use standard OpenFOAM finite volumes or high-order local reconstruction by LRE. Time integration is explicit forward Euler. The convergence figures are generated by `plot_convergence.m` from the data in `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAExplicit/results`

1 Manufactured FDA problem

The solved variables are the transmembrane potential V_m and the state variables u_1 , u_2 , and u_3 . In the manufactured problem,

$$\frac{\partial V_m}{\partial t} = \frac{1}{\chi C_m} \nabla \cdot (D \nabla V_m) - I_{\text{ion}}(V_m, u_1, u_2, u_3), \quad (1)$$

$$\frac{\partial u_1}{\partial t} = (u_1 + u_3 - V_m)^2 u_2^2 + \frac{1}{2} (u_1 + u_3 - V_m) u_2^2 (V_m - u_3), \quad (2)$$

$$\frac{\partial u_2}{\partial t} = -(u_1 + u_3 - V_m) u_2^3, \quad (3)$$

$$\frac{\partial u_3}{\partial t} = 0. \quad (4)$$

The ionic current used in the PDE is

$$I_{\text{ion}} = -\frac{1}{2} (u_1 + u_3 - V_m) u_2^2 (V_m - u_3) + \frac{\beta}{\chi C_m} (V_m - u_3), \quad (5)$$

where β is the exact Laplacian eigenvalue induced by the conductivity tensor and by the selected manufactured spatial mode.

For the two-dimensional runs reported here,

$$F(x, y) = \cos(\pi x) \cos(2\pi y), \quad (6)$$

$$G(x, y) = 1 + xy^2, \quad (7)$$

$$V_m^*(x, y, t) = \sqrt{1+t} F(x, y), \quad (8)$$

$$u_1^*(x, y, t) = (1+t)G(x, y) + V_m^*(x, y, t), \quad (9)$$

$$u_2^*(x, y, t) = \frac{1}{(1+t)\sqrt{G(x, y)}}, \quad (10)$$

$$u_3^* = 0. \quad (11)$$

The code also contains analogous one- and three-dimensional definitions.

2 Spatial discretization

The potential equation can use either the standard OpenFOAM finite-volume Laplacian or a high-order flux constructed from LRE. For the high-order option, local Taylor reconstructions are built from cell stencils. The reconstruction uses first derivatives for **p1**, first and second derivatives for **p2**, and up to third derivatives for **p3**. Face quadrature is used to integrate the diffusive flux, and cell quadrature is used to volume-average the nonlinear ionic current when high-order state reconstruction is enabled.

The switches `useHighOrder_Vm` and `useHighOrder_states` allow the potential and state variables to use different high-order levels. The expected spatial orders used in the plots are 2 for **p1**, 3 for **p2**, and 4 for **p3**. The standard non-HO option is shown with the second-order guide line.

3 Explicit time integration

At each time step the solver computes the ionic current and the Laplacian from known values at t^n , then applies forward Euler,

$$V_m^{n+1} = V_m^n + \Delta t \left[\frac{1}{\chi C_m} L_h(V_m^n) - I_{\text{ion},h}(V_m^n, u^n) \right], \quad (12)$$

$$u_i^{n+1} = u_i^n + \Delta t R_i(V_m^n, u^n), \quad i = 1, 2, 3. \quad (13)$$

The manufactured exact values are imposed at the relevant Dirichlet boundaries after each update.

4 Error measures and convergence rates

The solver writes cell-centred volume-weighted L_1 , L_2 , and L_∞ errors for V_m , u_1 , and u_2 . It also writes Gauss-reconstructed errors, denoted here by the superscript G , obtained by evaluating the reconstructed field at quadrature points. Pairwise rates use consecutive meshes,

$$p_{N_1, N_2} = \frac{\log(E_{N_1}/E_{N_2})}{\log(h_{N_1}/h_{N_2})}, \quad h_N = 1/N. \quad (14)$$

The summary table reports the all-mesh log-log fit of the cell-centred L_2 errors.

5 Observed orders

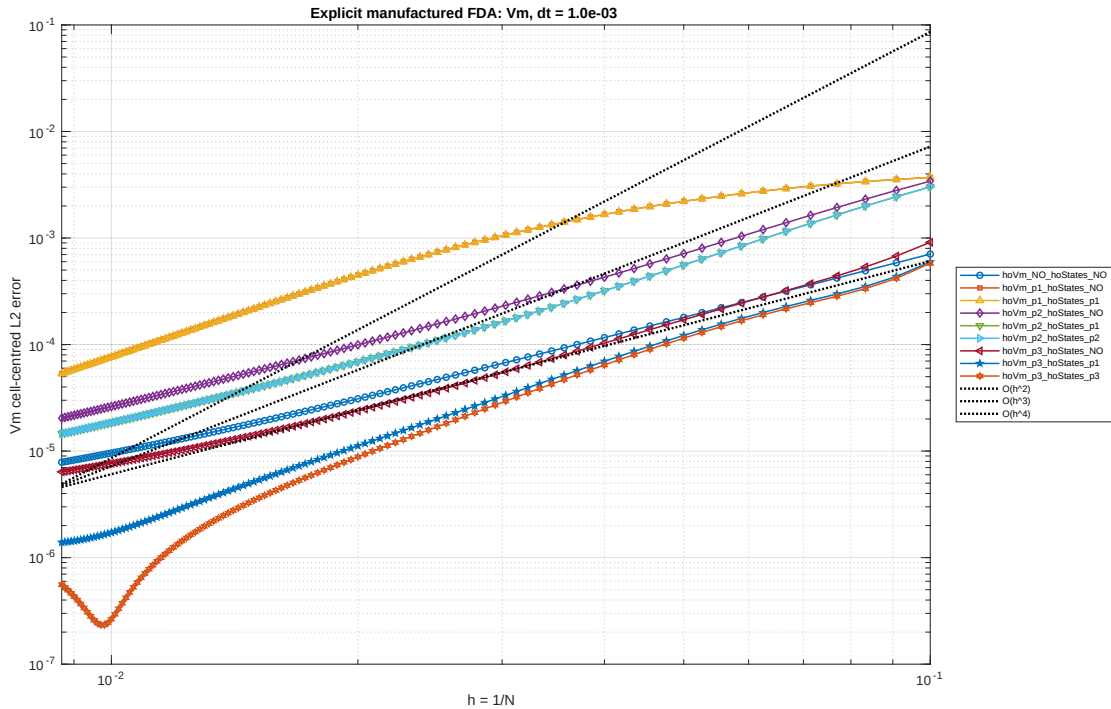
Table 1: Explicit solver: observed cell-centred L_2 convergence orders obtained from the all-mesh log-log fit. The expected orders use $p1 \mapsto 2$, $p2 \mapsto 3$, $p3 \mapsto 4$; the non-HO standard option is treated as second order.

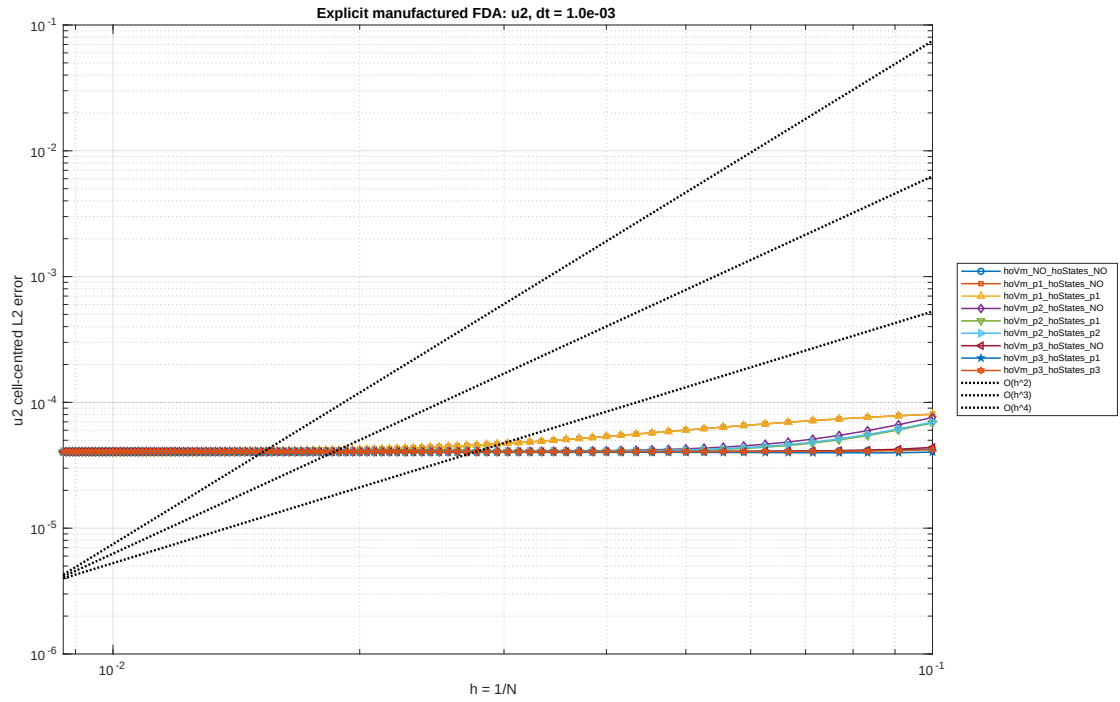
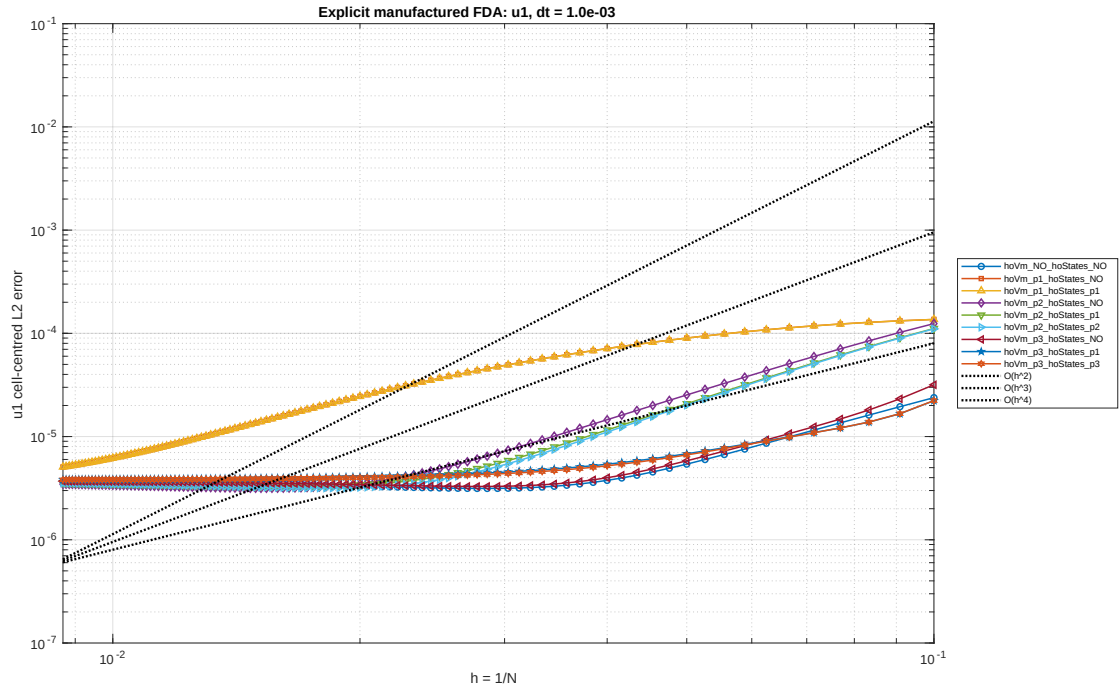
Case	Δt	Exp. V_m	V_m	Exp. states	u_1	u_2	V_m^G
hoVm_NO_hoStates_NO	10^{-3}	2	1.834	2	0.389	0.007	2.714
hoVm_NO_hoStates_NO	10^{-4}	2	1.979	2	1.885	0.264	2.859
hoVm_NO_hoStates_NO	10^{-5}	2	1.995	2	2.037	1.336	2.176
hoVm_p1_hoStates_NO	10^{-3}	2	1.992	2	1.586	0.255	2.145
hoVm_p1_hoStates_NO	10^{-4}	2	1.994	2	1.703	1.355	2.148
hoVm_p1_hoStates_NO	10^{-5}	2	1.994	2	1.703	1.705	2.149
hoVm_p1_hoStates_p1	10^{-3}	2	1.992	2	1.586	0.255	2.145

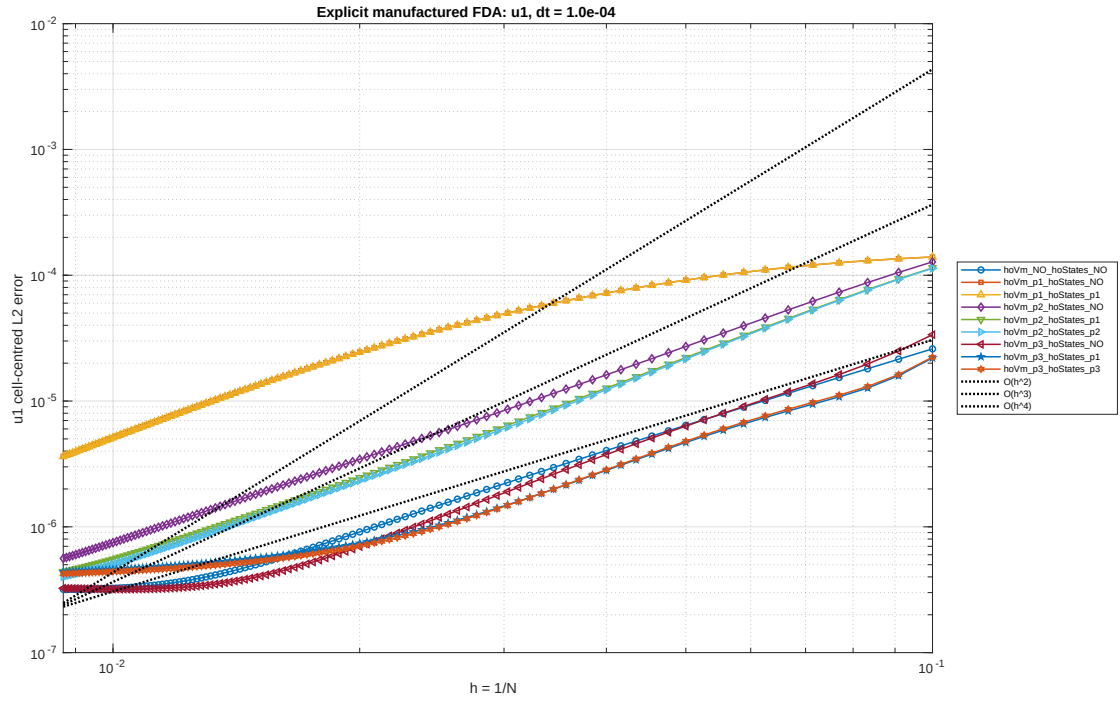
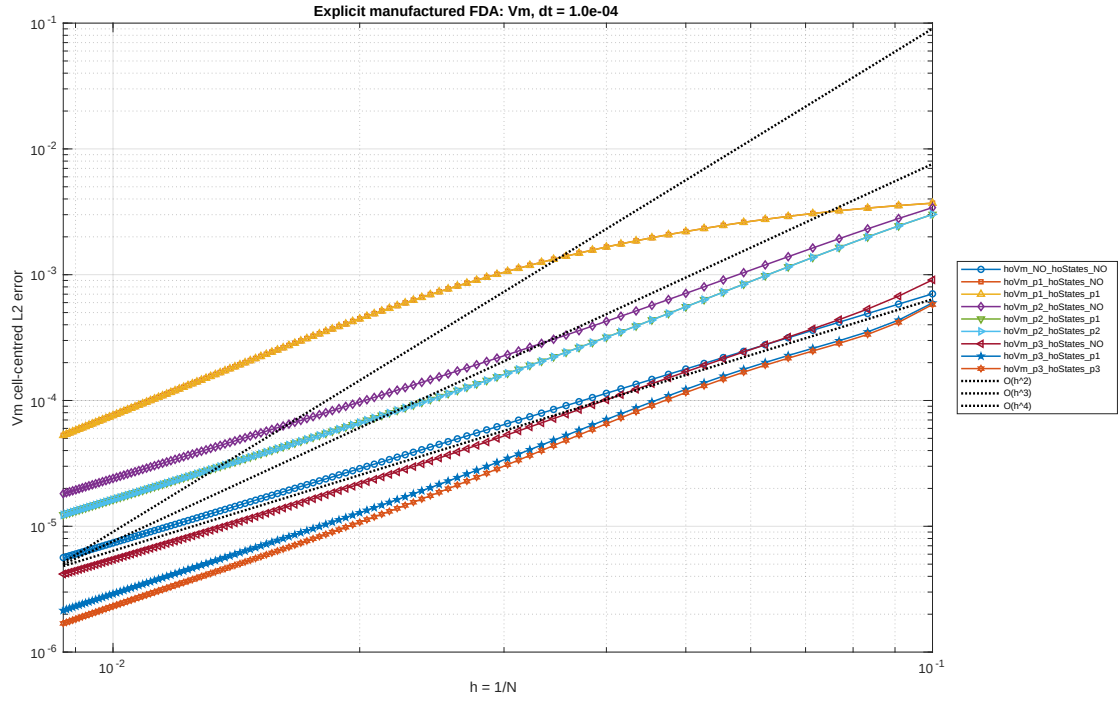
Case	Δt	Exp. V_m	V_m	Exp. states	u_1	u_2	V_m^G
hoVm_p1_hoStates_p1	10^{-4}	2	1.994	2	1.703	1.355	2.148
hoVm_p1_hoStates_p1	10^{-5}	2	1.994	2	1.703	1.705	2.149
hoVm_p2_hoStates_NO	10^{-3}	3	2.069	2	1.317	0.098	2.514
hoVm_p2_hoStates_NO	10^{-4}	3	2.118	2	2.229	0.877	2.557
hoVm_p2_hoStates_NO	10^{-5}	3	2.123	2	2.151	1.969	2.562
hoVm_p2_hoStates_p1	10^{-3}	3	2.139	2	1.172	0.071	2.658
hoVm_p2_hoStates_p1	10^{-4}	3	2.210	2	2.296	0.766	2.711
hoVm_p2_hoStates_p1	10^{-5}	3	2.217	2	2.245	1.973	2.717
hoVm_p2_hoStates_p2	10^{-3}	3	2.140	3	1.156	0.078	2.658
hoVm_p2_hoStates_p2	10^{-4}	3	2.211	3	2.336	0.773	2.712
hoVm_p2_hoStates_p2	10^{-5}	3	2.219	3	2.263	1.953	2.718
hoVm_p3_hoStates_NO	10^{-3}	4	1.953	2	0.437	0.009	2.382
hoVm_p3_hoStates_NO	10^{-4}	4	2.143	2	1.909	0.282	2.573
hoVm_p3_hoStates_NO	10^{-5}	4	2.166	2	2.235	1.349	2.596
hoVm_p3_hoStates_p1	10^{-3}	4	2.579	2	0.446	-0.007	3.168
hoVm_p3_hoStates_p1	10^{-4}	4	2.297	2	1.471	0.139	2.900
hoVm_p3_hoStates_p1	10^{-5}	4	2.263	2	2.214	1.164	2.865
hoVm_p3_hoStates_p3	10^{-3}	4	3.281	4	0.438	0.004	3.808
hoVm_p3_hoStates_p3	10^{-4}	4	2.393	4	1.506	0.181	3.037
hoVm_p3_hoStates_p3	10^{-5}	4	2.341	4	2.245	1.129	2.983

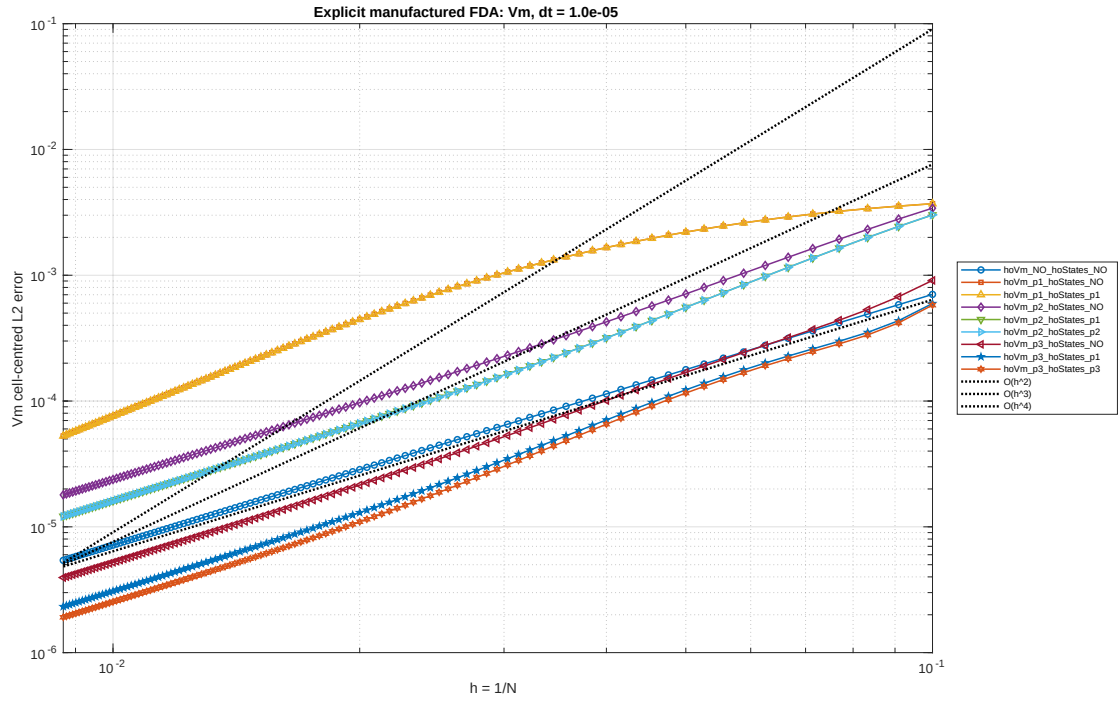
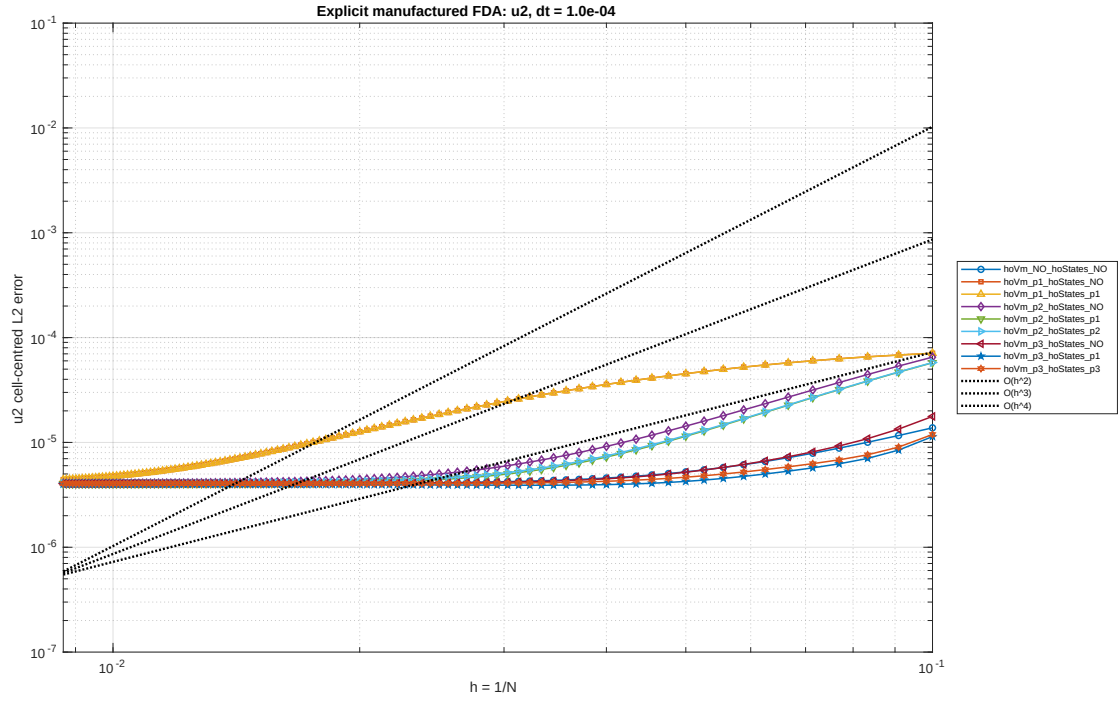
6 Convergence figures

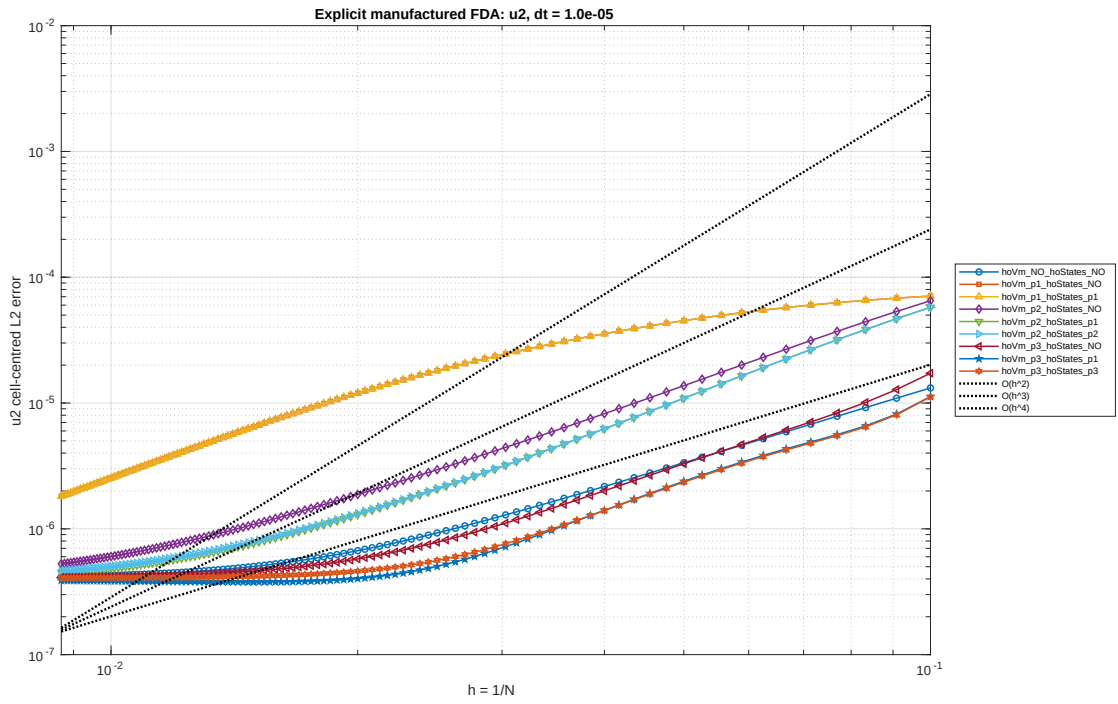
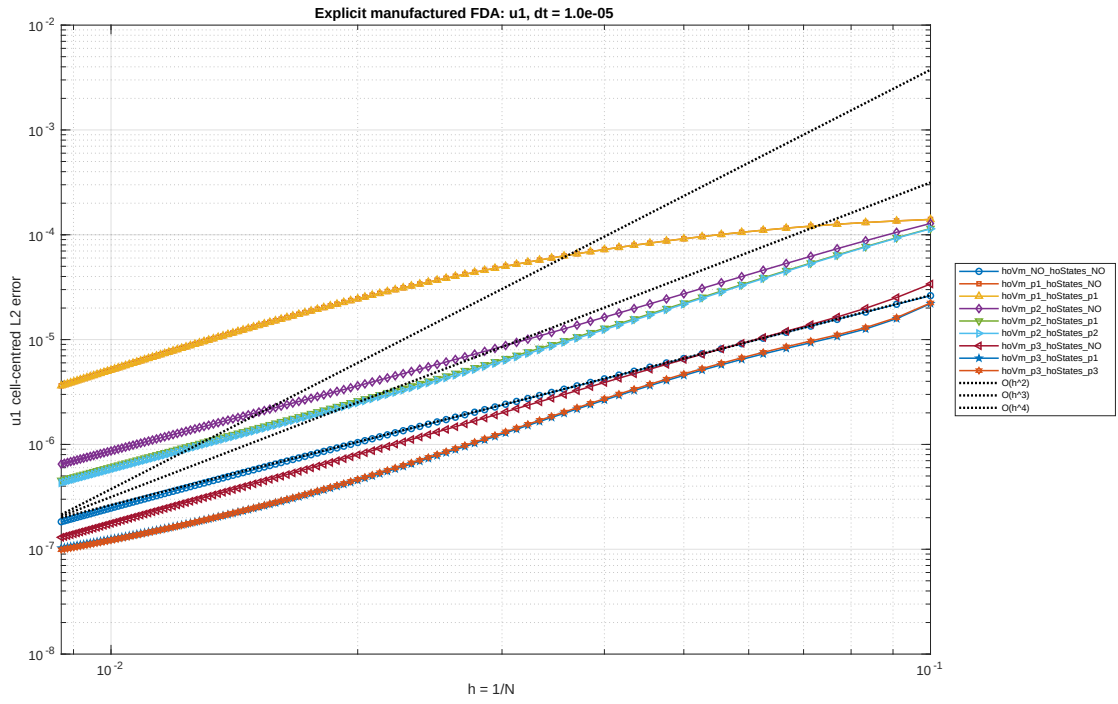
Run `plot_convergence.m` from this report directory to create the PDF and PNG figures. The plots compare all cases available in the explicit two-dimensional result directory for $\Delta t = 10^{-3}$, 10^{-4} , and 10^{-5} .











7 Notes

The explicit results may be affected by temporal error unless Δt is sufficiently small for the spatial error to dominate. When interpreting the plots, compare the three time-step levels and the reconstructed V_m^G column in Table [1](#) against the cell-centred V_m order.